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FUNCTION assign_triage_category(patient_data):

   // ----- Category 1: Immediate (life-threatening) -----
   IF (patient_data.GCS < 9) OR
      (patient_data.SBP > 160) OR
      (patient_data.pulse == 0) OR
      (patient_data.RR == 0) OR
      (patient_data.Temp > 41.0 OR patient_data.Temp < 32.0) OR
      (patient_data.Fio2 == 100 AND (patient_data.RR > 35 OR patient_data.pulse > 160)) OR   // high oxygen need
   THEN
       RETURN 1

   // ----- Category 2: Very Urgent (potential threat) -----
   IF (patient_data.GCS >= 9 AND patient_data.GCS <= 12) OR
      (patient_data.SBP >= 140 AND patient_data.SBP < 159) OR
      (patient_data.MAP < 65) OR
      (patient_data.pulse > 150 OR patient_data.pulse < 40) OR
      (patient_data.RR > 30 OR patient_data.RR < 8) OR
      (patient_data.Temp >= 39.0 AND patient_data.Temp < 41.0) OR
      (patient_data.Temp >= 35.0 AND patient_data.Temp < 36.0) OR
      (patient_data.Fio2 > 21 AND patient_data.Fio2 <= 60) OR   // needing oxygen
   THEN
       RETURN 2

   // ----- Category 3: Urgent (potentially serious) -----
   IF (patient_data.GCS >= 13 AND patient_data.GCS <= 15 ) OR
      (patient_data.SBP >= 120 AND patient_data.SBP < 139) OR
      (patient_data.DBP > 110) OR
      (patient_data.pulse >= 120 AND patient_data.pulse <= 150) OR
      (patient_data.RR >= 25 AND patient_data.RR <= 30) OR
      (patient_data.Temp >= 38.0 AND patient_data.Temp < 39.0) OR
      (patient_data.Fio2 > 21 AND patient_data.Fio2 <= 40) OR
   THEN
       RETURN 3

   // ----- Category 4: Standard (less urgent, stable) -----
   IF (patient_data.pulse >= 100 AND patient_data.pulse < 120) OR
      (patient_data.RR >= 20 AND patient_data.RR < 25) OR
      (patient_data.Temp >= 37.5 AND patient_data.Temp < 38.0) OR
      (patient_data.Fio2 == 21 AND (patient_data.RR < 20)) OR
   THEN
       RETURN 4

   // ----- Category 5: Non-urgent (minor) -----
   RETURN 5

END FUNCTION
```